History of Operating System

[redacted]

An operating system is a type of software that acts as an interface between the user and the hardware. It is responsible for handling various critical functions of the computer and utilizing resources very efficiently so the operating system is also known as a resource manager. The operating system also acts like a government because just as the government has authority over everything, similarly the operating system has authority over all resources. Various tasks that are handled by OS are file management, task management, garbage management, memory management, process management, disk management, I/O management, peripherals management, etc.

Generations of Operating Systems

- **1940s-1950s: Early Beginnings**
  - Computers operated without operating systems (OS).
  - Programs were manually loaded and run, one at a time.
  - The first operating system was introduced in 1956. It was a batch processing system (1956) that automated job handling.
- **1960s: Multiprogramming and Timesharing**
  - Introduction of multiprogramming to utilize CPU efficiently.
  - Timesharing systems, like CTSS (1961) and Multics (1969), allowed multiple users to interact with a single system.
- **1970s: Unix and Personal Computers**
  - Unix (1971) revolutionized OS design with simplicity, portability, and multitasking.
  - Personal computers emerged, leading to simpler OSs like CP/M (1974) and PC-DOS (1981).
- **1980s: GUI and Networking**
  - Graphical User Interfaces (GUIs) gained popularity with systems like Apple Macintosh (1984) and Microsoft Windows (1985).
  - Networking features, like TCP/IP in Unix, became essential.
- **1990s: Linux and Advanced GUIs**
  - Linux (1991) introduced open-source development.
  - Windows and Mac OS refined GUIs and gained widespread adoption.
- **2000s-Present: Mobility and Cloud**
  - Mobile OSs like iOS (2007) and Android (2008) dominate.

    - Cloud-based and virtualization technologies reshape computing, with OSs like Windows Server and Linux driving innovation.
- **AI Integration – (Ongoing)**
  With the growth of time, Artificial Intelligence came into picture. Operating system integrates features of AI technology like Siri, Google Assistant, and Alexa and became more powerful and efficient in many way. These AI features with operating system create a entire new feature like voice commands, predictive text, and personalized recommendations.

Note: The above mentioned OS basically tells how the OS evolved with the time by adding new features but it doesn't mean that only new generation OS are in use and previously OS system are not in use, according to the need, all these OS are still used in software industry.
Operating systems have evolved from basic program execution to complex ecosystems supporting diverse devices and users.

Function of Operating System

- Memory management
- Process management
- File management
- Device Management
- Deadlock Prevention
- Input/Output device management

History According to Types of Operating Systems

Operating Systems have evolved in past years. It went through several changes before getting its current form.

1. No OS – (0s to 1940s)

As we know that before 1940s, there was no use of OS . Earlier, people are lacking OS in their computer system so they had to manually type instructions for each tasks in machine language(0-1 based language) . And at that time , it was very hard for users to implement even a simple task. And it was very time consuming and also not user-friendly . Because not everyone had that much level of understanding to understand the machine language and it required a deep understanding.

2. Batch Processing Systems -(1950s)

With the growth of time, batch processing system came into the market .Now Users had facility to write their programs on punch cards and load it to the computer operator. And then operator make different batches of similar types of jobs and then serve the different batch(group of jobs) one by one to the CPU .CPU first executes jobs of one batch and them jump to the jobs of other batch in a sequence manner.

3. Multiprogramming Systems -(1960s and 1970s)

Multiprogramming was the first operating system where actual revolution began. It provide user facility to load the multiple program into the memory and provide a specific portion of memory to each program. When one program is waiting for any I/O operations (which take much time) at that time the OS give permission to CPU to switch from previous program to other program(which is first in ready queue) for continuous execution of program with interrupt.

4. Personal Computers Systems -(1970s)

Unix (1971) revolutionized OS design with simplicity, portability, and multitasking. Personal computers emerged, leading to simpler OSs like CP/M (1974) and PC-DOS (1981).

5. Introduction of GUI -(1980s)

With the growth of time, Graphical User Interfaces (GUIs) came. First time OS became more user-friendly and changed the way of people to interact with computer. GUI provides computer system visual elements which made user's interaction with computer more comfortable and user-friendly. User can just click on visual elements rather than typing commands. Here are some feature of GUI in Microsoft's windows icons, menus and windows.

6. Networked Systems – (1990s)

At 1980s, the craze of computer networks at it's peak .A special type of Operating Systems needed to manage the network communication. The OS like Novell NetWare and Windows NT were developed to manage network communication which provide users facility to work in collaborative environment and made file sharing and remote access very easy.

7. Mobile Operating Systems – (2000s)

Invention of smartphones create a big revolution in software industry, To handle the operation of smartphones , a special type of operating systems were developed. Some of them are : iOS and Android etc. These operating systems were optimized with the time and became more powerful.

8. AI Integration – (2010s to ongoing)

With the growth of time, Artificial intelligence came into picture. Operating system integrates features of AI technology like Siri, Google Assistant, and Alexa and became more powerful and efficient in many way. These AI features with operating system create a entire new feature like voice commands, predictive text, and personalized recommendations.

Advantages of Operating System

- Operating System manages external and internal devices for example, printers, scanners, and other.
- Operating System provides interfaces and drivers for proper communication between system and hardware devices.
- Allows multiple applications to run simultaneously.

- Manages the execution of processes, ensuring that the system remains responsive.
- Organizes and manages files on storage devices.
- Operating system allocates resources to various applications and ensures their efficient utilization.

Disadvantages of Operating System

- If an error occurred in your operating system, then there may be a chance that your data may not be recovered therefore always have a backup of your data.
- Threats and viruses can attack our operating system at any time, making it challenging for the OS to keep the system protected from these dangers.
- For learning about new operating system can be a time-consuming and challenging, Specially for those who using particular Operating system for example switching from Windows OS to Linux is difficult.
- Keeping an operating system up-to-date requires regular maintenance, which can be time-consuming.
- Operating systems consume system resources, including CPU, memory, and storage, which can affect the performance of other applications.

An operating system (OS) is a type of system software that manages a computer's hardware and software resources:

An operating system (OS) is system software that manages computer hardware and software resources, and provides common services for computer programs.

Difference between System Software and Operating System

Software is like a set of instructions that tell a computer what to do. There are two main types application software and system software. Application software is extra programs you can add to your computer to do specific things, like write documents or play games. You don't need these to use your computer, but they help you do particular tasks. System software is the basic software that runs your computer. It manages all the parts of the computer and makes sure everything works together. The most important system software is called an operating system. It helps you interact with

your computer and use all its features. Without system software, your computer wouldn't work at all.

What is System Software?

System Software is a main part of any computer system. It consists of programs that manage and control the computer's hardware components. Acting as a bridge between the user, hardware, and application software, system software ensures the smooth operation of the entire computer system. Its primary role is to coordinate various hardware elements and provide a stable environment for other software to run. System software handles essential tasks like managing memory, processing data, and controlling input/output operations.

Taking care of these fundamental functions, allows users to interact with the computer more easily and enables application software to perform specific tasks effectively. In essence, system software forms the foundation upon which all other software relies, making it an indispensable part of modern computing.

What is an Operating System?

The operating system is a fundamental piece of software that serves as the core of a computer system. It acts as an intermediary between the user's computer hardware and application software. When the computer is turned on the operating system is the first program to start initiated by BIOS (Basic Input/output System). Its primary role is to manage the computer resources including hardware components, memory, and storage. The operating system has a user-friendly environment that allows people to interact with the computer, run applications, and utilize various hardware features. It handles important tasks such as allocating system resources, managing files, and controlling input and output operations.

By taking care of these functions the operating system creates a stable platform for other software to run on making it an indispensable component of any computer system. The operating system acts as the central coordinator ensuring all parts of the computer work together smoothly and efficiently.

Difference Between System Software and Operating System

System Software	Operating System
The software which manages the resources and makes the interaction	An operating system is a software that communicates with your computer

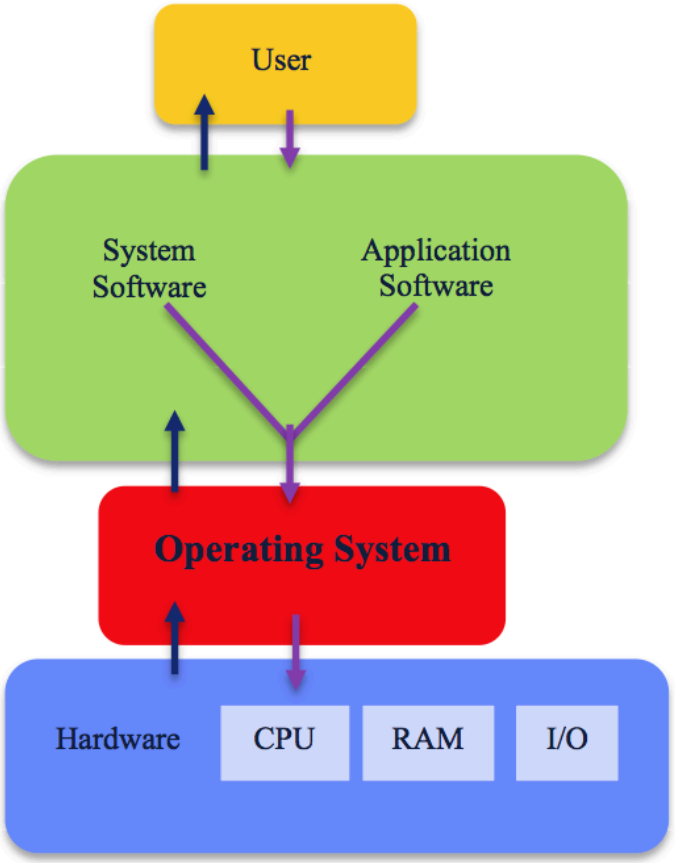
System Software	Operating System
between a user and machine possible is system software.	hardware and provides a place to run the application.
System software manages the system.	Operating System manages system as well as system software.
Types of System Software- • Operating Systems- MS Windows, Mac OS, Linux, etc. • Device Drivers- Display drivers, USB drivers, etc. • Firmware- BIOS, Embedded Systems, etc. • Language Translators- Compiler, Interpreter, Assembler • Utility- Antivirus software, WinRAR, etc.	**Types of Operating System-** • Batch Operating System • Multiprocessing Operating System • Multiprogramming Operating System • Multi Tasking Operating System • Time Sharing Operating System • Real-Time Operating System • Distributed Operating System • Network Operating System
It's run only when required.	It runs all the time.
It loads in the main memory whenever required.	It resides in the main memory all the time while the system is on.
It is loaded by the operating system.	It resides in the main memory all the time while the system is on.
Examples of system software are MacOS, Android, and Microsoft windows.	Examples of OS are Windows, OS X, and Linux.

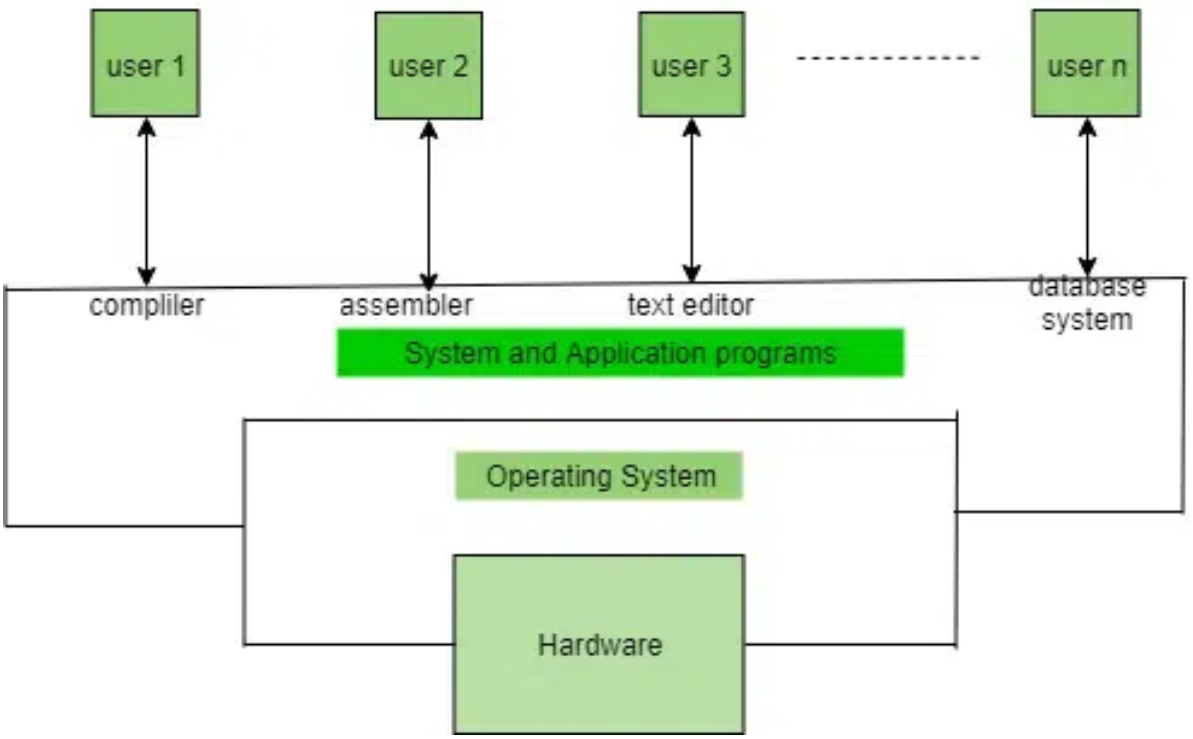
Major Difference between System Software and Operating System

- **Scope:** Operating system is a type of system software, but system software is not limited to operating system only. System software includes various other programs that help to manage and run the computer system, such as device drivers, language translators, utility programs, and more.
- **Function:** Operating system is the main program that manages and controls all the other software and hardware components of the computer. It provides a platform for other software applications to run on

top of it. System software, on the other hand, provides tools and services to help developers create and manage software applications.

- **Importance:** Operating system is an essential component of any computer system. Without an operating system, a computer cannot function. System software, on the other hand, is important but not essential. Some applications can run without system software, but they may not be able to utilize the full power of the computer system.
- **Complexity:** Operating systems are usually more complex than other types of system software. They have to manage hardware resources, provide security, and support various software applications. System software, on the other hand, can be relatively simple in comparison, depending on its purpose.

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